

Ai — Final Assignment (theory)

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Topic: PEAS Framework and Task Environment Analysis

Selected (Option B): A Personalized News Feed Recommender

Submitted to: Sir Rajesh Kumar

Task 1: PEAS Specification

The PEAS (Performance, Environment, Actuators, Sensors) framework for the Personalized News Feed Recommender is specified below:

1. Performance Measure:

Success is measured using four key engagement and satisfaction metrics:

- **Click-Through Rate & Watch Time:** Measures user interest and attention retention.
- **User Retention & Daily Active Users:** Measures long-term engagement and platform relevance.
- **Explicit Feedback Ratio:** Compares positive feedback (likes, shares, comments) with negative feedback (reports or dislikes).
- **Churn Rate:** Measures the percentage of users who stop using the platform over time.

2. Environment:

The environment includes online content, a user's past activity, current context (such as time and device), and content being published by other creators.

3. Actuators:

- **Content Rendering UI:** Displays recommended posts, videos, or articles to users.
- **Push Notification Engine:** Sends alerts for relevant content recommendations.
- **Ad Insertion API:** Places targeted sponsored content within the feed.

4. Sensors:

- **Click and Tap Loggers:** Track content selections, profile visits, and link clicks.
- **Watch-Duration Monitors:** Measure how long users view specific content.
- **Interaction Buttons (Likes/Shares/Comments):** Collect explicit user feedback.
- **Scroll Speed Tracker:** Measures user interest based on scrolling behavior.

Task 2: Environment Classification

- **Fully Observable vs. Partially Observable:** Partially Observable

Justification: The agent cannot fully know a user's mood, preferences, or true intent.

- **Single agent vs. Multi agent:** Multi-agent

Justification: Multiple users, creators, and ad systems continuously affect the environment.

- **Episodic vs. Sequential:** Sequential

Justification: Each recommendation affects future interactions and recommendations.

- **Static vs. Dynamic:** Dynamic

Justification: Content and user preferences change continuously over time.

- **Discrete vs. Continuous:** Continuous

Justification: User activity is measured using continuous values like watch time and interest scores.

Task 3: Critical Analysis

1. Greatest Technical Challenge

The Fully Observable vs. Partially Observable dimension poses the greatest technical challenge. Because a user's true intent and mood are hidden, the system must rely on indirect signals (clicks, watch time) that can easily be misinterpreted, such as an accidental tap. Advanced probabilistic models are therefore required to manage this uncertainty.

2. Utility Function and Trade-offs

Utility Function: $U = (0.6 \times \text{Predicted_Watch_Time}) + (0.3 \times \text{Content_Novelty}) - (0.3 \times \text{Ad_Fatigue_Penalty})$

This balances engagement, content freshness, and ad annoyance. If the watch-time weight is doubled, the algorithm would aggressively favor short-term, clickbait, or polarizing content to maximize screen time, risking filter bubbles, and lower long-term user satisfaction.

Structured Representation: in valid JSON format:

```
{
  "system_name": "Personalized News Feed Recommender",
  "peas": {
    "performance_measures": [
      "Click_Through_Rate",
      "User_Retention",
      "Explicit_Feedback_Ratio",
      "Churn_Rate"
    ],
    "environment": "Digital feed ecosystem, user interaction history, global content database",
    "actuators": [
      "Content_Rendering_UI",
```

```

        "Push_Notification_Engine",
        "Ad_Insertion_API"
    ],
    "sensors": [
        "Click_and_Tap_Loggers",
        "Dwell_Time_Monitors",
        "Interaction_Buttons",
        "Scroll_Speed_Tracker"
    ]
},
"environment_classification": {
    "observability": {
        "choice": "Partially Observable",
        "justification": "The agent cannot fully observe a user's intrinsic mood, offline preferences, or intent behind skipping content."
    },
    "agents": {
        "choice": "Multi-agent",
        "justification": "Multiple users, ad systems, and creators interact simultaneously, altering the global ecosystem."
    },
    "determinism": {
        "choice": "Stochastic",
        "justification": "Human behavior is inherently probabilistic and susceptible to external distractions."
    },
    "episodic_sequential": {
        "choice": "Sequential",
        "justification": "Each item shown alters the context and influences subsequent user interaction probabilities."
    },
    "static_dynamic": {
        "choice": "Dynamic",
        "justification": "The content pool grows constantly and user interests change, requiring real-time adaptations."
    },
    "discrete_continuous": {
        "choice": "Continuous",
        "justification": "Watch times, scroll velocities, and preference embeddings exist on continuous scales."
    }
},
"utility_function": "U = (0.6 * Predicted_Watch_Time) + (0.3 * Content_Novelty) - (0.3 * Ad_Fatigue_Penalty)"
}

```